

Package ‘DebarcodeR’

February 28, 2026

Title Automated Demultiplexing for Fluorescent Cell Barcoding

Version 1.1.0

Description DebarcodeR is an R package for demultiplexing fluorescent cell barcoded (FCB) flow cytometry data. Fluorescent cell barcoding is a technique for implementing pooled sample multiplexing in fluorescence flow cytometry using amine reactive dyes to label each sample with unique barcodes by varying the concentration of one or more dyes. Samples can then be pooled, stained, and acquired in a single tube, which reduces instrument time and reagent consumption and improves data robustness. Following acquisition, the data must be 'debarcoded' (also known as deconvolution or demultiplexing) which traditionally has required manual biaxial gating. DebarcodeR implements an algorithm for automating this demultiplexing process which improves reproducibility and enables high throughput data processing. For more information about DebarcodeR, see our manuscript and for more information about FCB, see Krutzik et al. 2006.

biocViews FlowCytometry, Preprocessing, SingleCell, CellBasedAssays, Clustering, OneChannel

URL <https://github.com/cytolab/DebarcodeR>

BugReports <https://github.com/cytolab/DebarcodeR/issues>

Depends R (>= 4.5.0)

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Encoding UTF-8

LazyData false

Imports flowCore,
pracma,
MASS,
scales,
mixsmsn,
sn,
classInt,
earth,
dplyr,
ggplot2,
ggforce,
ggnewscale,
janitor,

matrixStats,
mclust,
data.table,
methods

Suggests testthat (>= 3.0.0),

knitr,
rmarkdown,
BiocStyle,
flowWorkspace,
shiny (>= 1.8.0),
bslib (>= 0.6.0),
bsicons

VignetteBuilder knitr

Config/testthat/edition 3

RoxygenNote 7.3.3

Collate 'apply_platemap.R'

'fcbFlowFrame.R'
'assign_fcbFlowFrame.R'
'assign_fcbFlowSet.R'
'cluster_fcbFlowFrame.R'
'cluster_fcbFlowSet.R'
'deskew_fcbFlowFrame.R'
'deskew_fcbFlowSet.R'
'doRegressConstrained.R'
'em_optimize.R'
'fcbFlowSet.R'
'getAssignments.R'
'jurkat-data.R'
'morphology_corr_earth.R'
'morphology_corr_knignenburg.R'
'morphology_corr_lm.R'
'morphology_corr_master.R'
'plot-fcbFlowFrame-assignments.R'
'run_debarcoder.R'
'selectDenseScatterArea.R'
'split_fcbFlowFrame.R'
'utils-barcode-slots.R'
'utils-validation.R'

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apply_platemap

Apply a plate map to a debarcoded flowSet

Description

Joins barcoding level assignments to human-readable well names, updates pData, and renames each flowFrame to <prefix>_<well>.

Usage

```
apply_platemap(fcFlowSet, platemap, drop0 = FALSE, prefix = NA)
```

Arguments

fcFlowSet	An fcFlowSet produced by split .
platemap	A data.frame mapping barcoding level combinations to well names. Must contain a well column plus one column per barcoding channel (using cleaned names as produced by <code>janitor::clean_names</code>).
drop0	Logical. If TRUE, drop unassigned (level 0) cells. Default FALSE concatenates all unassigned cells into a single "Unassigned" flowFrame.
prefix	Character, prefix for output FCS filenames. If NA (default), the prefix is taken from the \$FILENAME keyword in each FCS file.

Value

An fcFlowSet with updated sampleNames and pData, where each sample is named <prefix>_<well>.

See Also

[getAssignments](#) and [split](#) for the preceding steps

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
library(flowCore)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Blue-A")
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Orange-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Orange-A", levels = 6)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Orange-A")
assignments <- getAssignments(fcb)
fcbfs <- split(fcb, assignments)
myplatemap <- data.frame(
  pacific_blue_a = as.character(rep(1:8, times = 6)),
  pacific_orange_a = as.character(rep(1:6, each = 8)),
  well = paste0(
    rep(LETTERS[1:8], times = 6),
    formatC(rep(1:6, each = 8), width = 2, flag = "0")
  )
)
fcbfs <- apply_platemap(fcbfs, myplatemap, prefix = "Jurkat_FCB")
sampleNames(fcbfs)
```

assign_fcbFlowFrame *Assign cells to barcoding levels*

Description

Applies likelihood and ambiguity cutoffs to the probability matrix from [cluster_fcbFlowFrame](#) (or [em_optimize](#)) to discretely assign each cell to a barcoding level. Unassigned cells are given a level of 0 rather than being discarded.

Usage

```
assign_fcbFlowFrame(
  fcbFlowFrame,
  channel,
  likelihoodcut = 8,
  ambiguitycut = 0.02
)
```

Arguments

fcbFlowFrame	An fcbFlowFrame with completed deskewing and clustering for the target channel.
channel	The name (string) of the channel to assign, or "wells" to assign from the multivariate em_optimize result.
likelihoodcut	Numeric. Cells less than 1/likelihoodcut as likely as the most likely cell in their population are left unassigned (default 8).
ambiguitycut	Numeric between 0 and 1. Cells whose highest population probability is below 1 - ambiguitycut are left unassigned. E.g., 0.02 unassigns cells with more than a 2% chance of belonging to a different population (default 0.02).

Value

An fcbFlowFrame with the assignment slot populated for the specified channel. Assigned level values are stored as a character vector; 0 indicates unassigned.

See Also

[cluster_fcbFlowFrame](#) for the preceding step, [em_optimize](#) for multivariate refinement before assigning "wells", [getAssignments](#) to extract the result

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Blue-A")
```

assign_fcbFlowSet	<i>Assign cells to barcoding levels for a fcbFlowSet</i>
-------------------	--

Description

Applies [assign_fcbFlowFrame](#) to each frame in an fcbFlowSet.

Usage

```
assign_fcbFlowSet(fcbFlowSet, channel, likelihoodcut = 8, ambiguitycut = 0.02)
```

Arguments

fcFlowSet	An fcFlowSet object with deskewing and clustering completed.
channel	The name (string) of the channel to assign.
likelihoodcut	Numeric, a likelihood cutoff for discarding unlikely cells. Cells less than 1/likelihoodcut as likely as the most likely cell from that population are left unassigned (default 8).
ambiguitycut	Numeric from 0 to 1, threshold below which to leave cells unassigned. E.g., 0.02 unassigns cells with more than 2% chance of originating from another population (default 0.02).

Value

An fcFlowSet object with assignment slots populated.

See Also

[assign_fcbFlowFrame](#) for single-frame processing, [cluster_fcbFlowSet](#) for the preceding step

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
library(flowCore)
fcbfs <- fcFlowSet(flowSet(list(A = jurkatFCB)))
fcbfs <- deskew_fcbFlowSet(fcbfs,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcbfs <- cluster_fcbFlowSet(fcbfs,
  channel = "Pacific Blue-A",
  levels = 8, opt = "fisher"
)
fcbfs <- assign_fcbFlowSet(fcbfs, channel = "Pacific Blue-A")
```

cluster_fcbFlowFrame *Defines populations on barcoded datasets*

Description

This function allows you to calculate the probability of a cell originating from a given population using either gaussian mixture modeling or jenks natural breaks classification

Usage

```
cluster_fcbFlowFrame(
  fcbFlowFrame,
  channel,
  levels,
  opt = "mixture",
  dist = NULL,
```

```

    subsample = 3000,
    trim = 0,
    ret.model = TRUE,
    manbreaks = NULL,
    updateProgress = NULL
  )

```

Arguments

<code>fcbFlowFrame</code>	An <code>fcbFlowFrame</code> object post deskewing (at least one channel in the barcodes slot).
<code>channel</code>	The name (string) of the channel to be clustered.
<code>levels</code>	Integer, the number of barcoding intensities present in the channel.
<code>opt</code>	String: "mixture" (default) for Gaussian mixture modeling, "fisher" for Fisher-Jenks natural breaks, or "manual.breaks" for manually specified breakpoints.
<code>dist</code>	String, one of <code>c("Normal", "Skew.normal", "Tdist")</code> , passed to <code>mixsmsn::smsn.mix</code> . Ignored for <code>opt = "fisher"</code> and <code>opt = "manual.breaks"</code> .
<code>subsample</code>	Integer, number of cells to subsample for model fitting (default 3000).
<code>trim</code>	Numeric between 0 and 1; trims the upper and lower extremes to exclude outliers. E.g., <code>trim = 0.01</code> excludes the most extreme 1% of cells (default 0).
<code>ret.model</code>	Logical, whether to retain the fitted model (default TRUE).
<code>manbreaks</code>	Numeric vector of length <code>levels + 1</code> specifying manual breakpoints; required when <code>opt = "manual.breaks"</code> .
<code>updateProgress</code>	Callback function used in a Shiny context to report progress (default NULL).

Value

An `fcbFlowFrame` with the clustering slot populated for the specified channel. The slot contains a probability matrix with `nrow = ncells` and `ncol = levels`. For Gaussian mixture modeling, probabilities reflect each cell's probability of originating from each level under the fitted distribution. For Jenks/manual breaks, probabilities are estimated empirically from per-level histograms.

See Also

[deskew_fcbFlowFrame](#) for the preceding pipeline step, [assign_fcbFlowFrame](#) for the next pipeline step, [cluster_fcbFlowSet](#) to process an entire `flowSet`

Examples

```

data(jurkatFCB)
data(jurkatFCB_std)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)

```

cluster_fcbFlowSet	<i>Defines populations on barcoded datasets</i>
--------------------	---

Description

This function allows you to calculate the probability of a cell originating from a given population using either gaussian mixture modeling or jenks natural breaks classification

Usage

```
cluster_fcbFlowSet(
  fcbFlowSet,
  channel,
  levels,
  opt = "mixture",
  dist = NULL,
  subsample = 10000,
  trim = 0,
  ret.model = TRUE,
  updateProgress = NULL
)
```

Arguments

fcbFlowSet	An fcbFlowSet object post deskewing (at least one channel in the barcodes slot).
channel	The name (string) of the channel to be clustered.
levels	Integer, the number of barcoding intensities present in the channel.
opt	String: "mixture" (default) for Gaussian mixture modeling, or "fisher" for Fisher-Jenks natural breaks.
dist	String, one of c("Normal", "Skew.normal", "Tdist"), passed to <code>mixsmsn::smsn.mix</code> . Ignored for <code>opt = "fisher"</code> .
subsample	Integer, number of cells to subsample for model fitting (default 10000).
trim	Numeric between 0 and 1; trims the upper and lower extremes to exclude outliers (default 0).
ret.model	Logical, whether to retain the fitted model (default TRUE).
updateProgress	Callback function used in a Shiny context to report progress (default NULL).

Value

An fcbFlowSet with clustering slots populated for all frames.

See Also

[cluster_fcbFlowFrame](#) for single-frame processing, [deskew_fcbFlowSet](#) for the preceding step, [assign_fcbFlowSet](#) for the next step

Examples

```

data(jurkatFCB)
data(jurkatFCB_std)
library(flowCore)
fcbfs <- fcbFlowSet(flowSet(list(A = jurkatFCB)))
fcbfs <- deskew_fcbFlowSet(fcbfs,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcbfs <- cluster_fcbFlowSet(fcbfs,
  channel = "Pacific Blue-A",
  levels = 8, opt = "fisher"
)

```

deskew_fcbFlowFrame	<i>Corrects morphology based on scatter or uptake control for fcbFlowFrame</i>
---------------------	--

Description

Corrects morphology based on scatter or uptake control for fcbFlowFrame

Usage

```

deskew_fcbFlowFrame(
  fcbFlowFrame,
  uptake = NULL,
  channel,
  method = "earth",
  predictors = c("FSC-A", "SSC-A"),
  subsample = 20000,
  ret.model = TRUE,
  verbose = FALSE,
  updateProgress = NULL,
  ...
)

```

Arguments

fcbFlowFrame	An fcbFlowFrame, flowFrame, or cytoframe object, post compensation and pre-processing.
uptake	A flowFrame or cytoframe to use as the uptake (external standard) control. If NULL (default), the barcoded sample itself is used.
channel	The name (string) of the channel to be corrected (use original column name, e.g. "Pacific Blue-A").
method	The morphology correction method: "earth" (default), "lm", or "knijnenburg".
predictors	Character vector of channel names for the regression model.
subsample	Integer, number of cells to subsample for model fitting (default 20000).
ret.model	Logical, whether to retain the fitted model (default TRUE).

verbose Logical, whether to print progress messages (default FALSE).
 updateProgress Callback function for Shiny progress updates (default NULL).
 ... Additional arguments passed to the morphology correction method.

Value

An fcbFlowFrame with barcode slot populated for the specified channel.

See Also

[cluster_fcbFlowFrame](#) for the next pipeline step, [deskew_fcbFlowSet](#) to process an entire flowSet, [morphology_corr.earth](#) for details on the default method

Examples

```

data(jurkatFCB)
data(jurkatFCB_std)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(
  fcb,
  uptake      = jurkatFCB_std,
  channel     = "Pacific Blue-A",
  predictors  = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb
  
```

deskew_fcbFlowSet	<i>Corrects morphology based on scatter or uptake control for fcbFlowSet</i>
-------------------	--

Description

Applies [deskew_fcbFlowFrame](#) to each frame in an fcbFlowSet.

Usage

```

deskew_fcbFlowSet(
  fcbFlowSet,
  uptake = NULL,
  channel,
  method = "earth",
  predictors = c("FSC-A", "SSC-A"),
  subsample = 20000,
  ret.model = TRUE,
  verbose = FALSE,
  updateProgress = NULL,
  ...
)
  
```

Arguments

fcfFlowSet	An fcfFlowSet, flowSet, or cytoset object.
uptake	A flowFrame to use as the uptake control (default NULL).
channel	The name (string) of the channel to be corrected.
method	The morphology correction method: "earth" (default), "lm", or "knijnenburg".
predictors	Character vector of channel names for the regression model.
subsample	Integer, number of cells to subsample (default 20000).
ret.model	Logical, whether to retain fitted models (default TRUE).
verbose	Logical, whether to print progress messages (default FALSE).
updateProgress	Callback function for Shiny progress updates (default NULL).
...	Additional arguments passed to the morphology correction method.

Value

An fcfFlowSet with barcode slots populated for the specified channel.

See Also

[deskew_fcbFlowFrame](#) for single-frame processing, [cluster_fcbFlowSet](#) for the next pipeline step

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
library(flowCore)
fcbfs <- fcbFlowSet(flowSet(list(A = jurkatFCB)))
fcbfs <- deskew_fcbFlowSet(
  fcbfs,
  uptake      = jurkatFCB_std,
  channel     = "Pacific Blue-A",
  predictors  = c("FSC-A", "SSC-A", "APC-H7-A")
)
```

Description

Refines univariate cluster assignments using a multivariate Gaussian mixture model that accounts for correlations between barcoding channels. Requires that all channels have been deskewed, clustered, and assigned.

Usage

```
em_optimize(
  fcbFlowFrame,
  modelName = "VEE",
  subsample = 50000,
  verbose = TRUE,
  niter = 1,
  shrinkage = 0.05
)
```

Arguments

fcbFlowFrame	An fcbFlowFrame with completed deskewing, clustering, and assignment for all barcoding channels.
modelName	Character, mclust model name (default "VEE").
subsample	Numeric, number of cells to subsample during M step (default 50000).
verbose	Logical, print progress messages (default TRUE).
niter	Integer, number of EM iterations (default 1).
shrinkage	Numeric, fraction of simulated data for shrinkage regularization (default 0.05). Set to 0 to disable.

Value

An fcbFlowFrame with a "wells" entry in the barcodes slot containing multivariate clustering probabilities. Pass "wells" as the channel argument to [assign_fcbFlowFrame](#) to convert these probabilities into discrete well assignments.

See Also

[assign_fcbFlowFrame](#) for the assignment step (use channel = "wells"), [cluster_fcbFlowFrame](#) for the preceding univariate clustering

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
fcb <- fcbFlowFrame(jurkatFCB)
# Deskew, cluster, and assign both barcoding channels
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Blue-A")
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Orange-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Orange-A", levels = 6)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Orange-A")
# EM optimize across channels
```

```
fcb <- em_optimize(fcb, niter = 3)
fcb <- assign_fcbFlowFrame(fcb,
  channel = "wells",
  likelihoodcut = 12, ambiguitycut = 0.05
)
```

fcbFlowFrame-class	<i>fcbFlowFrame class</i>
--------------------	---------------------------

Description

A container for barcoded flow cytometry data, extending flowCore's flowFrame class with a barcodes slot that stores deskewing, clustering, and assignment results.

Constructor function to create an fcbFlowFrame from a flowFrame or cytoframe. If a cytoframe is provided (from the flowWorkspace package), it is automatically converted to a flowFrame first.

Usage

```
fcbFlowFrame(x, barcodes = list())
```

```
fcbFlowFrame(x, barcodes = list())
```

Arguments

x	A flowFrame or cytoframe object.
barcodes	A list of barcode data (default: empty list).

Value

An object of class fcbFlowFrame.

Slots

barcodes	A named list storing debarcoding results per channel. Each element is named by channel and contains sublists for deskewing, clustering, and assignment.
----------	---

See Also

[deskew_fcbFlowFrame](#) to begin the debarcoding pipeline, `as(x, "flowFrame")` for coercion back to flowFrame

Examples

```
data(jurkatFCB)
fcb <- fcbFlowFrame(jurkatFCB)
fcb
```

fcbFlowSet	Create an fcbFlowSet
------------	----------------------

Description

Constructor function to create an fcbFlowSet from a flowSet or cytoset. If a cytoset is provided (from the flowWorkspace package), it is automatically converted to a flowSet first.

Usage

```
fcbFlowSet(x)
```

Arguments

x A flowSet or cytoset object.

Value

An object of class fcbFlowSet.

See Also

[split](#) to produce an fcbFlowSet from an assigned fcbFlowFrame, `as(x, "flowSet")` for coercion back to flowSet

Examples

```
data(jurkatFCB)
library(flowCore)
fs <- flowSet(list(sample1 = jurkatFCB))
fcbfs <- fcbFlowSet(fs)
```

fcbFlowSet-class	fcbFlowSet class
------------------	------------------

Description

A container for debarcoded flow cytometry data, extending flowCore's flowSet class. Contains a collection of flowFrames, one per debarcoded sample.

Examples

```
data(jurkatFCB)
library(flowCore)
fcbfs <- fcbFlowSet(flowSet(list(sample1 = jurkatFCB)))
class(fcbfs)
```

getAssignments	<i>Extract assignments from an fcbFlowFrame or fcbFlowSet</i>
----------------	---

Description

Returns the discrete cell-level assignments as a named list of factors, one per barcoding channel. The result is suitable for passing directly to [split](#) to generate one flowFrame per barcode combination.

Usage

```
getAssignments(x, platemap = NULL, simplify = FALSE)
```

Arguments

x	An fcbFlowFrame or fcbFlowSet that has been assigned.
platemap	Data frame, a lookup table from barcoding levels to well assignments (optional; not yet implemented).
simplify	Logical, return a single factor rather than a list (not yet implemented; default FALSE).

Value

A named list of factors, one per barcoding channel that has been assigned. The "wells" entry (from [em_optimize](#)) is excluded — use channel = "wells" in [assign_fcbFlowFrame](#) to include multivariate well labels.

See Also

[assign_fcbFlowFrame](#) to generate assignments, [split](#) to split data by the returned factors, [apply_platemap](#) to map level combinations to well names

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Blue-A")
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Orange-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Orange-A", levels = 6)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Orange-A")
assignments <- getAssignments(fcb)
str(assignments)
```

get_barcode_data	<i>Get data from the barcodes slot</i>
------------------	--

Description

Retrieves pipeline results stored in the barcodes slot of an fcbFlowFrame. Preferred over direct @barcodes access.

Usage

```
get_barcode_data(fcbFF, channel, step, field = NULL)
```

Arguments

fcbFF	An fcbFlowFrame object.
channel	Channel name.
step	Pipeline step: "deskewing", "clustering", or "assignment".
field	Optional field within the step (e.g., "values", "probabilities", "model").

Value

The requested data, or NULL if the field does not exist.

See Also

[fcbFlowFrame](#), [deskew_fcbFlowFrame](#), [cluster_fcbFlowFrame](#), [assign_fcbFlowFrame](#)

Examples

```
data(jurkatFCB)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A")
)
deskewed <- get_barcode_data(fcb, "Pacific Blue-A", "deskewing", "values")
head(deskewed)
```

jurkatFCB	<i>48 Jurkat Samples Barcoded using Fluorescent Cell Barcoding</i>
-----------	--

Description

Dataset consisting of 48 Jurkat samples stained with combinations of Pacific Orange (6 levels), Pacific Blue (8 levels), and Alexa Fluor 750 (1 level, on channel APC-H7-A) for a total of 48 wells. Samples were acquired as separate FCS files and concatenated into a single file, with row and col channels added to record each cell's original sample of origin.

Usage

```
data(jurkatFCB)
```

Format

An object of class "flowFrame" with 24,000 cells and 16 channels; see [flowFrame-class](#).

References

Reisman et al. (2021) [doi:10.1002/cyto.a.24363](https://doi.org/10.1002/cyto.a.24363)

See Also

[jurkatFCB_std](#) for the companion external standard

Examples

```
data(jurkatFCB)
jurkatFCB
```

jurkatFCB_std

External Standard for the jurkatFCB Dataset

Description

A single-well subset of [jurkatFCB](#) (row 1, column 1) intended for use as an external standard (uptake control) in the DebarcodeR tutorial. This population was stained with a single level of each barcoding dye, so it provides a direct measurement of dye uptake per cell unconfounded by multi-level mixing.

Usage

```
data(jurkatFCB_std)
```

Format

An object of class "flowFrame" with 500 cells and 16 channels; see [flowFrame-class](#).

References

Reisman et al. (2021) [doi:10.1002/cyto.a.24363](https://doi.org/10.1002/cyto.a.24363)

See Also

[jurkatFCB](#) for the full pooled dataset, [deskew_fcbFlowFrame](#) for the uptake parameter that accepts this object

Examples

```
data(jurkatFCB_std)
jurkatFCB_std
```

plot.fcbFlowFrame	<i>Plot debarcoding assignments</i>
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Description

Plot debarcoding assignments

Usage

```
## S3 method for class 'fcbFlowFrame'
plot(x, ..., plot = "assignments", seed = 1)
```

Arguments

x	An fcbFlowFrame with completed assignments.
...	Additional arguments (not used).
plot	Character, plot type: "assignments" (default), "density", "chull", or "data".
seed	Deprecated; no longer used.

Value

A ggplot2 object.

Examples

```
data(jurkatFCB)
data(jurkatFCB_std)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Blue-A")
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Orange-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Orange-A", levels = 6)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Orange-A")
plot(fcb)
```

run_debarcoder

*Launch the DebarcodeR Shiny GUI***Description**

Opens an interactive Shiny application that walks through the DebarcodeR debarcoding pipeline step by step: load data, configure channels, deskew, cluster, assign, and export. As the user tunes parameters interactively, the app generates a reproducible R script that can be copied into an analysis pipeline.

Usage

```
run_debarcoder(data = NULL, uptake = NULL, launch.browser = TRUE)
```

Arguments

data Optional preprocessed flowFrame, cytoframe, or fcbFlowFrame. When provided the file-upload step is skipped and the app starts with channel configuration. Data should already be compensated, transformed, and gated to the target population.

uptake Optional flowFrame to use as the external standard for deskewing. Must have the same transformations applied as data.

launch.browser Logical; passed to [runApp](#). Default TRUE opens the app in the system browser.

Value

Called for its side effect (launches a Shiny app). Returns the value of [runApp](#) invisibly.

Examples

```
if (interactive()) {
  run_debarcoder()

  # Or pass data from the R session:
  # run_debarcoder(data = my_gated_ff, uptake = my_std_ff)
}
```

show, fcbFlowFrame-method

*Show method for fcbFlowFrame***Description**

Prints a concise summary of the fcbFlowFrame including cell count, channel count, and the debarcoding state (which pipeline steps have been completed) for each barcoded channel.

Usage

```
## S4 method for signature 'fcbFlowFrame'
show(object)
```

Arguments

object An fcbFlowFrame object.

Value

Invisibly returns object.

Examples

```
data(jurkatFCB)
fcb <- fcbFlowFrame(jurkatFCB)
show(fcb)
```

split, fcbFlowFrame, list-method

Split an fcbFlowFrame into an fcbFlowSet by assignments

Description

Splits an fcbFlowFrame into separate flowFrames based on barcode assignments, returned as an fcbFlowSet with updated pData.

Usage

```
## S4 method for signature 'fcbFlowFrame,list'
split(x, f, drop = FALSE, prefix = NULL, flowSet = TRUE, merge.na = TRUE, ...)
```

Arguments

x An fcbFlowFrame object with completed assignments.

f A list of assignment factors (from [getAssignments](#)).

drop Not used.

prefix Not used.

flowSet Logical, whether to return a flowSet (default TRUE).

merge.na Not used.

... Additional arguments (not used).

Value

An fcbFlowSet with each flowFrame named by assignment level, and pData columns for each bar-coding channel.

See Also

[getAssignments](#) to generate f, [apply_platemap](#) to rename wells after splitting

Examples

```

data(jurkatFCB)
data(jurkatFCB_std)
library(flowCore)
fcb <- fcbFlowFrame(jurkatFCB)
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Blue-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Blue-A", levels = 8)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Blue-A")
fcb <- deskew_fcbFlowFrame(fcb,
  uptake = jurkatFCB_std,
  channel = "Pacific Orange-A",
  predictors = c("FSC-A", "SSC-A", "APC-H7-A")
)
fcb <- cluster_fcbFlowFrame(fcb, channel = "Pacific Orange-A", levels = 6)
fcb <- assign_fcbFlowFrame(fcb, channel = "Pacific Orange-A")
assignments <- getAssignments(fcb)
fcbfs <- split(fcb, assignments)
pData(fcbfs)

```

split, fcbFlowSet, list-method

Split an fcbFlowSet by assignments

Description

Split an fcbFlowSet by assignments

Usage

```

## S4 method for signature 'fcbFlowSet,list'
split(
  x,
  f,
  assign0 = FALSE,
  drop = FALSE,
  prefix = NULL,
  flowSet = TRUE,
  merge.na = TRUE,
  ...
)

```

Arguments

x	An fcbFlowSet object.
f	A list of assignment factors per sample.
assign0	Logical, whether to assign level 0 cells (default FALSE).
drop	Not used.

prefix	Not used.
flowSet	Logical, whether to return a flowSet (default TRUE).
merge.na	Not used.
...	Additional arguments (not used).

Value

An fcbFlowSet with each flowFrame named by the assignments.

Examples

```
# See split,fcbFlowFrame,list-method for a single-frame example
```

```
split,flowFrame,list-method
```

Split a flowFrame by a list of factors

Description

Split a flowFrame by a list of factors

Usage

```
## S4 method for signature 'flowFrame,list'
split(x, f, drop = FALSE, prefix = NULL, flowSet = TRUE, merge.na = TRUE, ...)
```

Arguments

x	A flowFrame object.
f	A list of factors for splitting.
drop	Not used.
prefix	Not used.
flowSet	Logical, whether to return a flowSet (default TRUE).
merge.na	Not used.
...	Additional arguments (not used).

Value

An fcbFlowSet with each flowFrame named by the combined factor levels.

Examples

```
# See split,fcbFlowFrame,list-method for a full example
```